CHAPTER 6

APPENDIX

In [129…

```
%matplotlib inline
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import os
from glob import glob
import seaborn as sns
```

In [130…

```
base_skin_dir = "/Users/Keerthi Vardhani/Desktop/python/Skin Lesion/skin_images"
skin_df = pd.read_csv(os.path.join(base_skin_dir, 'HAM10000_metadata.csv')) # load in th
skin_df.head()
```

Out[130]:

	lesion_id	image_id	dx	dx_type	age	sex	localization	dataset
0	HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp	vidir_modern
1	HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp	vidir_modern
2	HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp	vidir_modern
3	HAM_0002730	ISIC_0025661	bkl	histo	80.0	male	scalp	vidir_modern
4	HAM_0001466	ISIC_0031633	bkl	histo	75.0	male	ear	vidir_modern

In [136…

```
imageid_path_dict = {os.path.splitext(os.path.basename(x))[0]: x
                     for x in glob(os.path.join('/Users/Keerthi Vardhani/Desktop/python/

lesion_type_dict = {
    'nv': 'Melanocytic_nevi',
    'mel': 'melanoma',
    'bkl': 'Benign_keratosis-like_lesions',
    'bcc': 'Basal_cell_carcinoma',
    'akiec': 'Actinic_keratoses',
    'vasc': 'Vascular_lesions',
    'df': 'Dermatofibroma'
}

lesion_danger = {
    'nv': 0, # 0 for benign
    'mel': 1, # 1 for malignant
    'bkl': 0, # 0 for benign
    'bcc': 1, # 1 for malignant
    'akiec': 1, # 1 for malignant
    'vasc': 0,
    'df': 0
}
```

In [137…

```
skin_df["path"] = skin_df["image_id"].map(imageid_path_dict.get) # map image_id to the p
```

In [138…

```
skin_df["path"] = skin_df["image_id"].map(imageid_path_dict.get) # map image_id to the p
```

In [139…

```
skin_df["cell_type"] = skin_df["dx"].map(lesion_type_dict.get) # map dx to type of lesio
```

In [140…

```
skin_df.head()
```

Out[140]:

	lesion_id	image_id	dx	dx_type	age	sex	localization	dataset	path
0	HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin

									Le...
1	HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...
2	HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...
3	HAM_0002730	ISIC_0025661	bkl	histo	80.0	male	scalp	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...
4	HAM_0001466	ISIC_0031633	bkl	histo	75.0	male	ear	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...

In [141...

```
skin_df["Malignant"] = skin_df["dx"].map(lesion_danger.get)
```

In [142...

```
skin_df.head()
```

Out[142]:

	lesion_id	image_id	dx	dx_type	age	sex	localization	dataset	path
0	HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...
1	HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...
2	HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...
3	HAM_0002730	ISIC_0025661	bkl	histo	80.0	male	scalp	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...
4	HAM_0001466	ISIC_0031633	bkl	histo	75.0	male	ear	vidir_modern	/Users/Keerthi Vardhani/Desktop/python/Skin Le...

In [143...

```
skin_df["cell_type_idx"] = pd.Categorical(skin_df["cell_type"]).codes # give each cell t
```

In [144...

```
skin_df.sample(3)
```

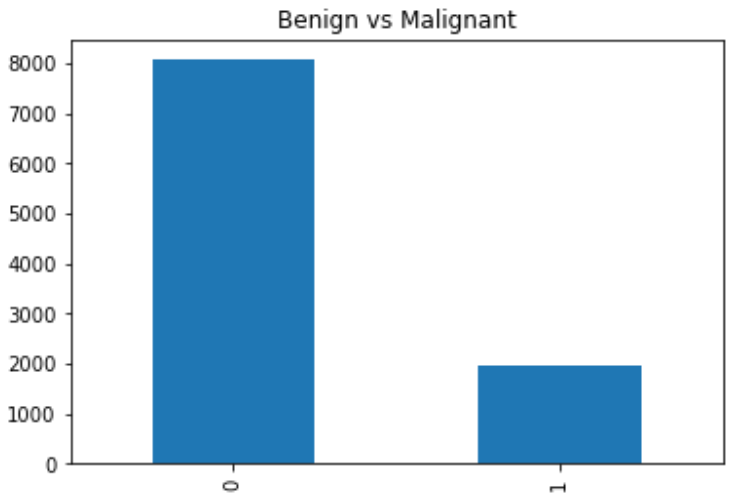
Out[144]:

	lesion_id	image_id	dx	dx_type	age	sex	localization	dataset	p
2863	HAM_0002017	ISIC_0025630	bcc	histo	60.0	female	back	rosendahl	/Users/Kee Vardhani/Desktop/python/
186	HAM_0004975	ISIC_0033778	bkl	histo	85.0	male	chest	vidir_modern	/Users/Kee Vardhani/Desktop/python/
7646	HAM_0007124	ISIC_0034311	nv	histo	55.0	male	back	vidir_modern	/Users/Kee Vardhani/Desktop/python/

In [145...

```
skin_df["Malignant"].value_counts().plot(kind="bar", title="Benign vs Malignant")
```

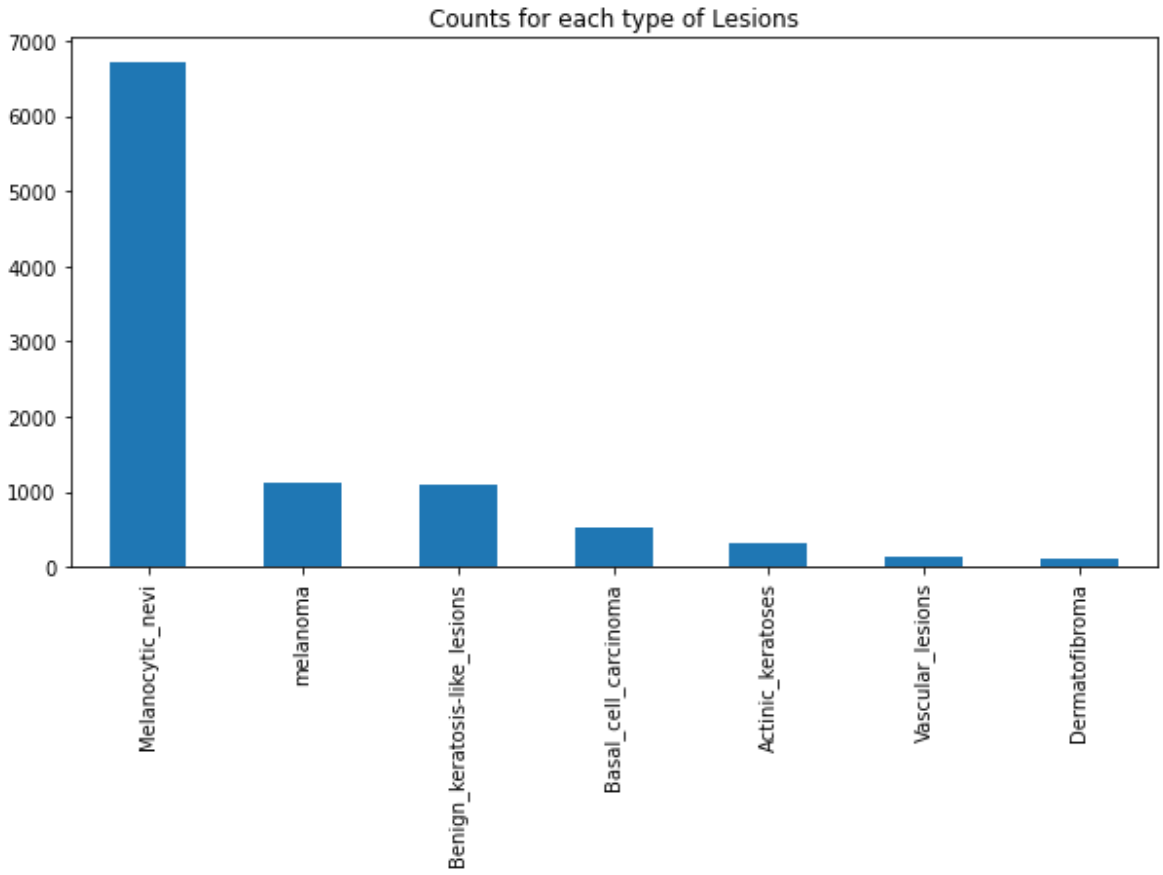
Out[145]: <AxesSubplot:title={'center':'Benign vs Malignant'}>



Most cases in our dataset are benign.

In [146…
```
fig, ax1 = plt.subplots(1,1,figsize=(10,5))
skin_df["cell_type"].value_counts().plot(kind="bar", ax=ax1, title="Counts for each type
```

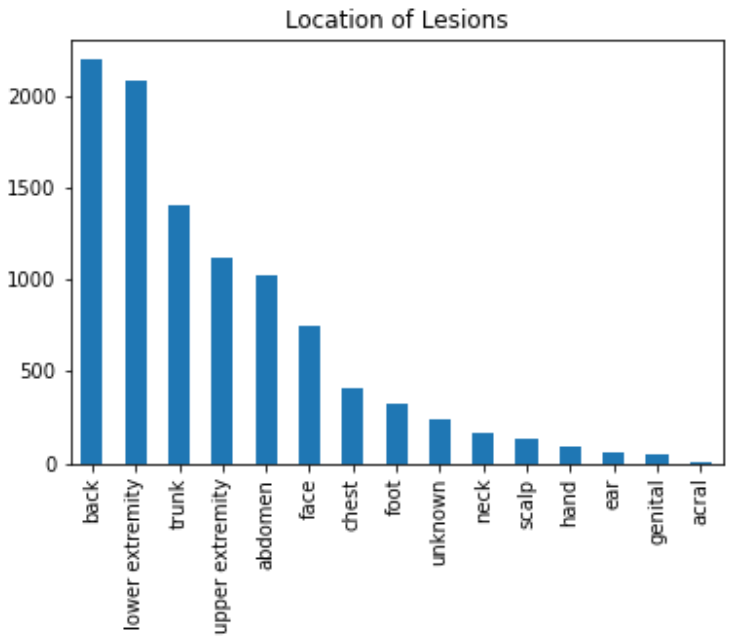
Out[146]: <AxesSubplot:title={'center':'Counts for each type of Lesions'}>



Our dataset is biased toward Melanocytic nevi. The cell_type with the second highest samples is the noctorious melanoma

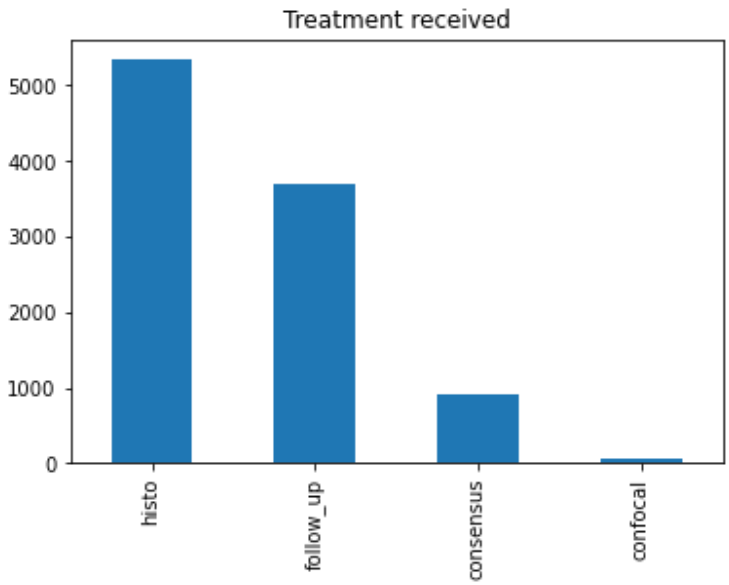
In [147…
```
# let's see where lesions are mostly located
skin_df["localization"].value_counts().plot(kind='bar', title="Location of Lesions")
```

Out[147]: <AxesSubplot:title={'center':'Location of Lesions'}>



In [148…
```
skin_df["dx_type"].value_counts().plot(kind='bar', title="Treatment received")
```

Out[148]:
```
<AxesSubplot:title={'center':'Treatment received'}>
```



Description for each dx_type:

histo: "Histopathologic diagnoses of excised lesions have been performed by specialized dermatopathologists."

follow_up: "If nevi monitored by digital dermatoscopy did not show any changes during 3 follow-up visits or 1.5 years we accepted this as evidence of biologic benignity. Only nevi, but no other benign diagnoses were labeled with this type of ground-truth because dermatologists usually do not monitor dermatofibromas, seborrheic keratoses, or vascular lesions."

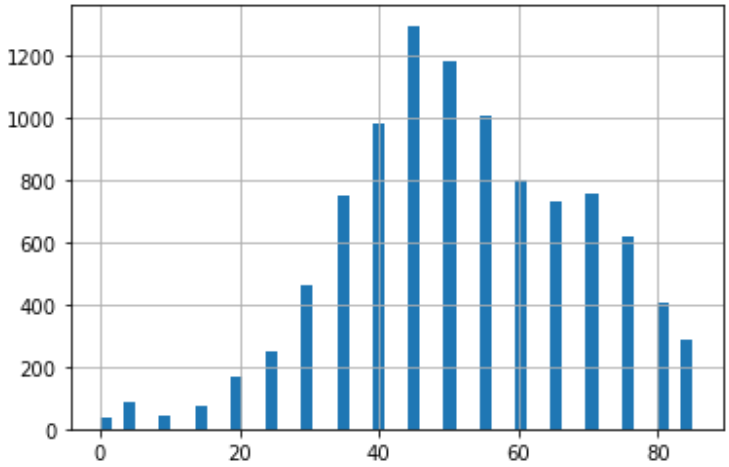
consensus: "For typical benign cases without histopathology or follow-up we provide an expertconsensus rating of authors PT and HK. We applied the consensus label only if both authors independently gave the same unequivocal benign diagnosis. Lesions with this type of ground-truth were usually photographed for educational reasons and did not need further follow-up or biopsy for confirmation."

confocal: "Reflectance confocal microscopy is an in-vivo imaging technique with a resolution at near-cellular level, and some facial benign keratoses were verified by this method."

Let's look at some characteristics of our patients

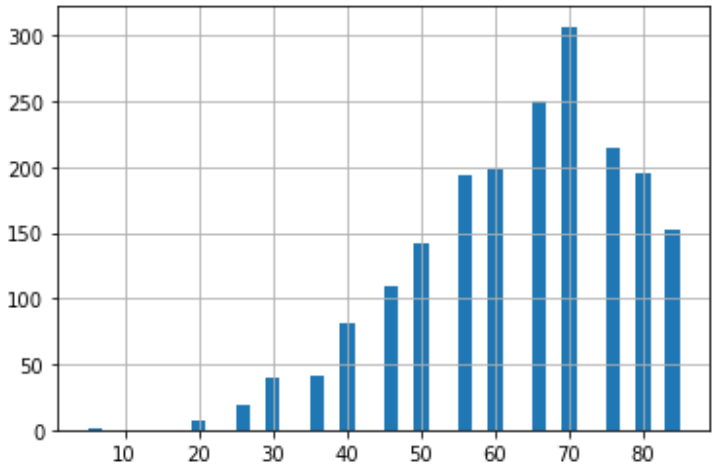
In [149…
```
skin_df["age"].hist(bins=50)
```

Out[149]:
```
<AxesSubplot:>
```



In [150…
```
skin_df[skin_df["Malignant"] == 1]["age"].hist(bins=40)
```

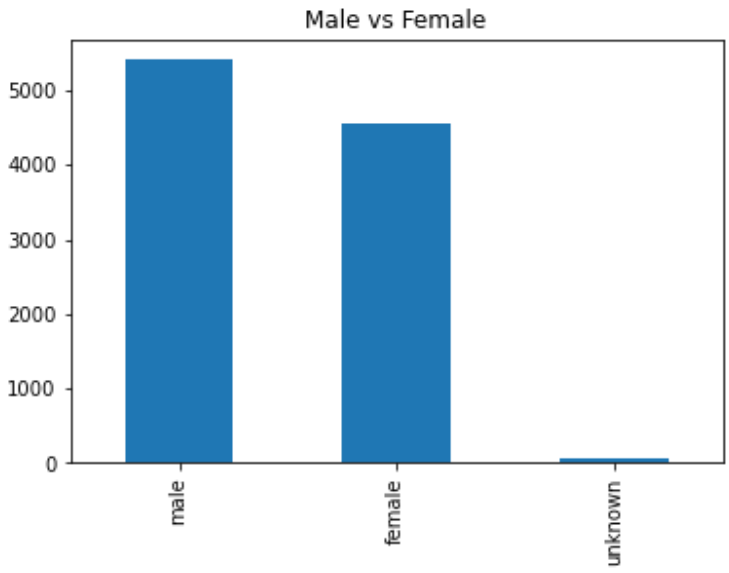
Out[150]:
```
<AxesSubplot:>
```



We can see that most of patients are above 30. But for the malignant cases, most patients are 50 and above, and 70s - year - old patients are the most present.

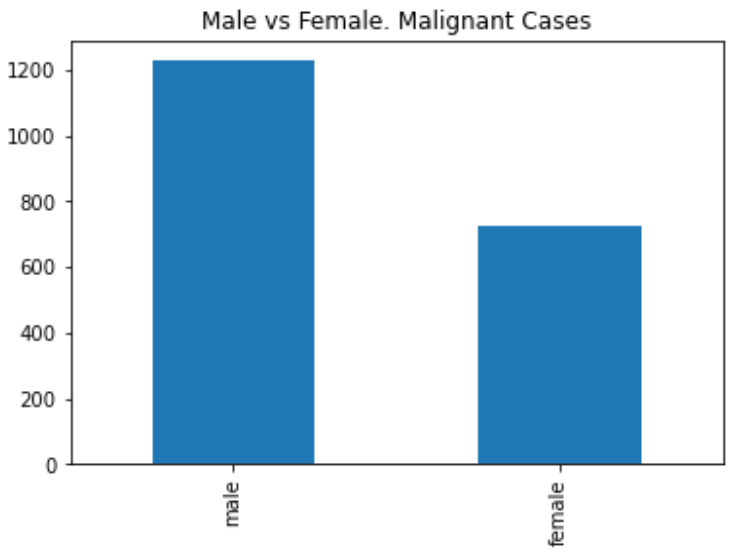
In [151…
```
skin_df["sex"].value_counts().plot(kind="bar", title="Male vs Female")
```

Out[151]:
```
<AxesSubplot:title={'center':'Male vs Female'}>
```



In [152…
```
skin_df[skin_df["Malignant"] == 1]["sex"].value_counts().plot(kind="bar", title="Male vs
```

Out[152]:
```
<AxesSubplot:title={'center':'Male vs Female. Malignant Cases'}>
```



We have more male patients than female patients in both general population and in malignant case. So far we haven't looked at our image yet. So let's now change our focus into how lesions in our dataset look like.

In [153…
```
from skimage.io import imread
```

In [154…
```
skin_df["image"] = skin_df["path"].map(imread) # read the image to array values
```

In [155…
```
skin_df.iloc[0]["image"] # here is a sample
```

Out[155]:
```
array([[[188, 147, 191],
        [186, 148, 189],
        [187, 150, 191],
        ...,
        [196, 155, 171],
        [197, 156, 170],
        [197, 157, 168]],

       [[186, 149, 193],
        [187, 152, 194],
        [189, 153, 191],
        ...,
```

```
         [194, 156, 169],
         [195, 159, 169],
         [192, 159, 168]],

        [[185, 148, 192],
         [189, 152, 195],
         [190, 153, 196],
         ...,
         [196, 155, 169],
         [198, 157, 171],
         [194, 156, 169]],

        ...,

        [[157, 124, 155],
         [156, 121, 154],
         [159, 124, 154],
         ...,
         [177, 146, 161],
         [176, 144, 159],
         [175, 141, 155]],

        [[155, 122, 151],
         [156, 123, 154],
         [156, 123, 152],
         ...,
         [178, 147, 163],
         [175, 144, 159],
         [175, 142, 159]],

        [[154, 119, 151],
         [153, 120, 149],
         [154, 121, 152],
         ...,
         [176, 147, 167],
         [175, 147, 161],
         [173, 143, 155]]], dtype=uint8)
```

In [156…

```python
# let's see what is the shape of each value in the image column
skin_df["image"].map(lambda x: x.shape).value_counts()
```

Out[156]:

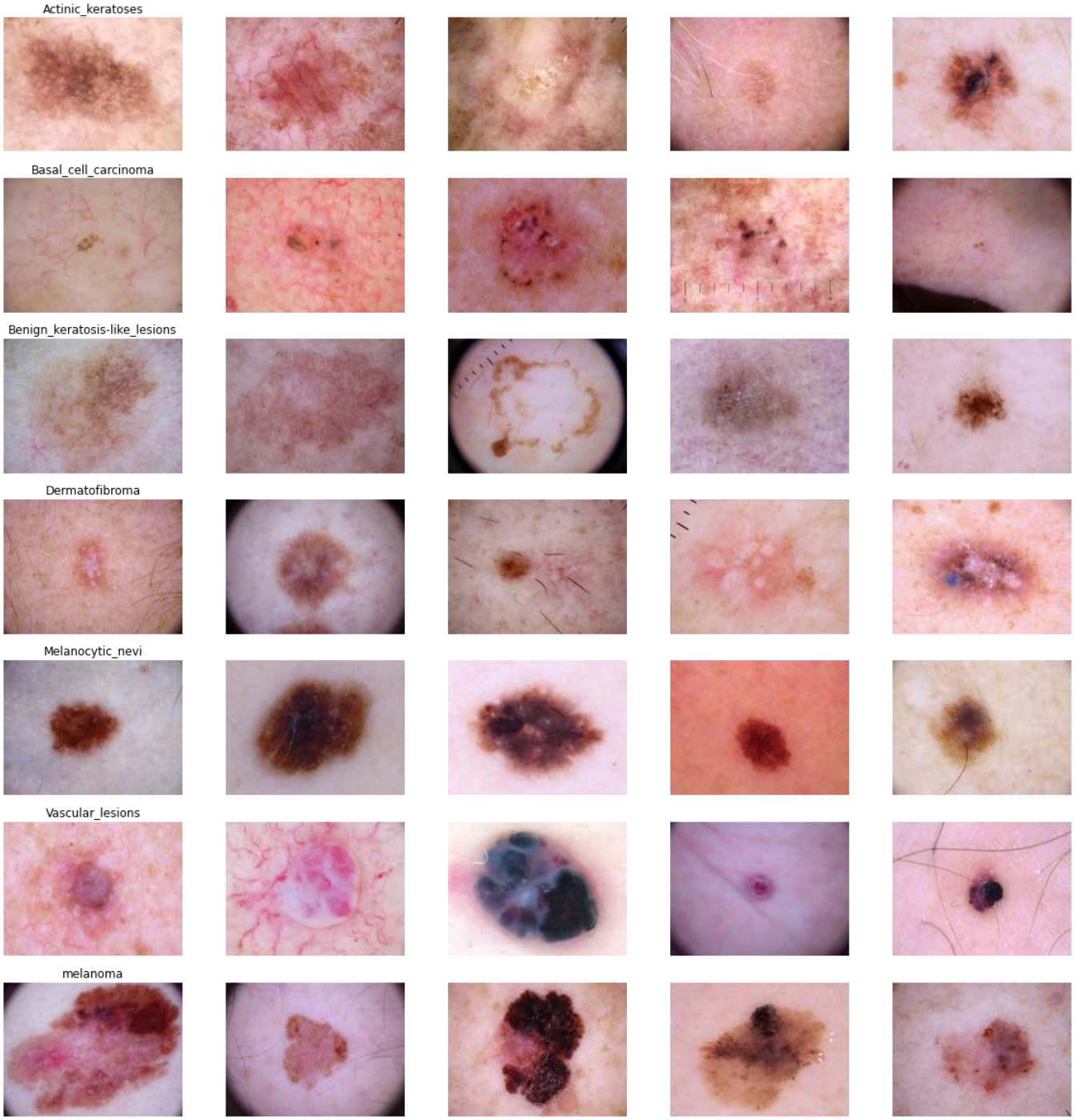
```
(450, 600, 3)    10015
Name: image, dtype: int64
```

In [157…

```python
# let's have a look at the image data

n_samples = 5 # choose 5 samples for each cell type
fig, m_axs = plt.subplots(7, n_samples, figsize=(4*n_samples, 3 * 7))

for n_axs, (type_name, type_rows) in zip(m_axs, skin_df.sort_values(["cell_type"]).group
    n_axs[0].set_title(type_name)
    for c_ax, (_, c_row) in zip(n_axs, type_rows.sample(n_samples, random_state=0).iterr
        c_ax.imshow(c_row["image"])
        c_ax.axis("off")
fig.savefig("category_samples.png", dpi=300)
```



Based on these images, it is still very hard for non-experts to know which is which.

Get Average Color Information

Here we get and normalize all of the color channel information

The shape of the image array is (450, 600, 3). 3 are the 3 chanels: Red, Blue and Green! Taking the mean across axis=(0,1) gives the mean for each 3 channels.

In [158…

```
# create a pandas dataframe to store mean value of Red, Blue and Green for each picture
rgb_info_df = skin_df.apply(lambda x: pd.Series({'{}_mean'.format(k): v for k, v
                                                 in zip(["Red", "Blue", "Green"],
                                                        np.mean(x["image"], (0, 1)))}),


gray_col_vec = rgb_info_df.apply(lambda x: np.mean(x), 1) # take the mean value across c
```

```
for c_col in rgb_info_df.columns:
    rgb_info_df[c_col] = rgb_info_df[c_col]/gray_col_vec
rgb_info_df["Gray_mean"] = gray_col_vec
rgb_info_df.sample(3)
```

Out[158]:

	Red_mean	Blue_mean	Green_mean	Gray_mean
222	1.239232	0.842864	0.917904	158.066477
7722	1.170062	0.923832	0.906106	182.885278
1597	1.169155	0.899264	0.931581	132.603205

In [159…

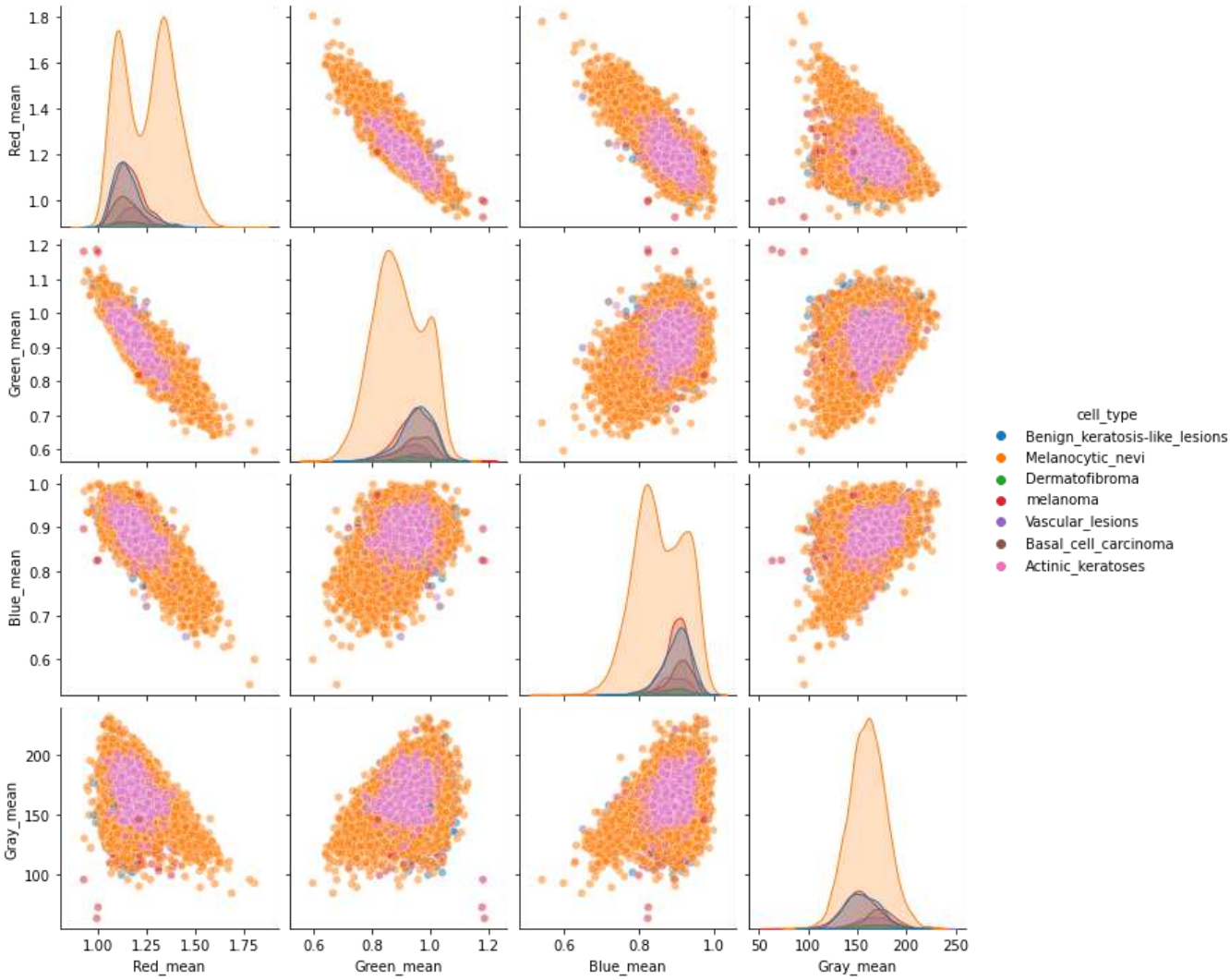
```
for c_col in rgb_info_df.columns:
    skin_df[c_col] = rgb_info_df[c_col].values
```

In [160…

```
# let's draw a plot showing the distribution of different cell types over colors!
sns.pairplot(skin_df[["Red_mean", "Green_mean", "Blue_mean", "Gray_mean", "cell_type"]],
             hue="cell_type", plot_kws = {"alpha": 0.5})
```

Out[160]:

```
<seaborn.axisgrid.PairGrid at 0x28505bef550>
```



Changes in cell type appearance as values in color chanel changes

In this section, I am doing an analysis on how each cell type looks like when each color channel values changes. E.g. the first 5 images demonstrate how cell Actinic Keratoses appearance changes as the values in

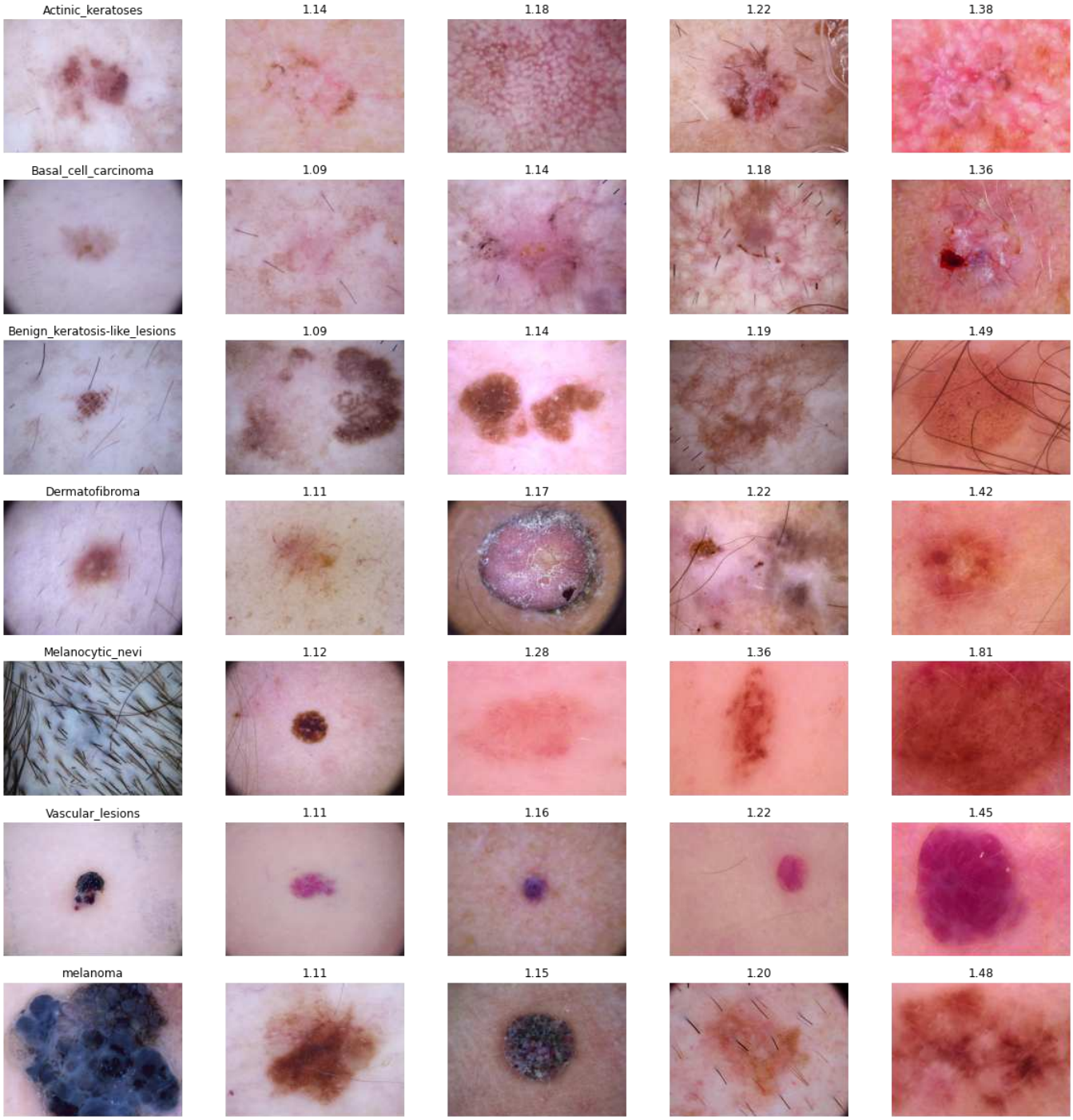
red channel gets bigger.

In [161…

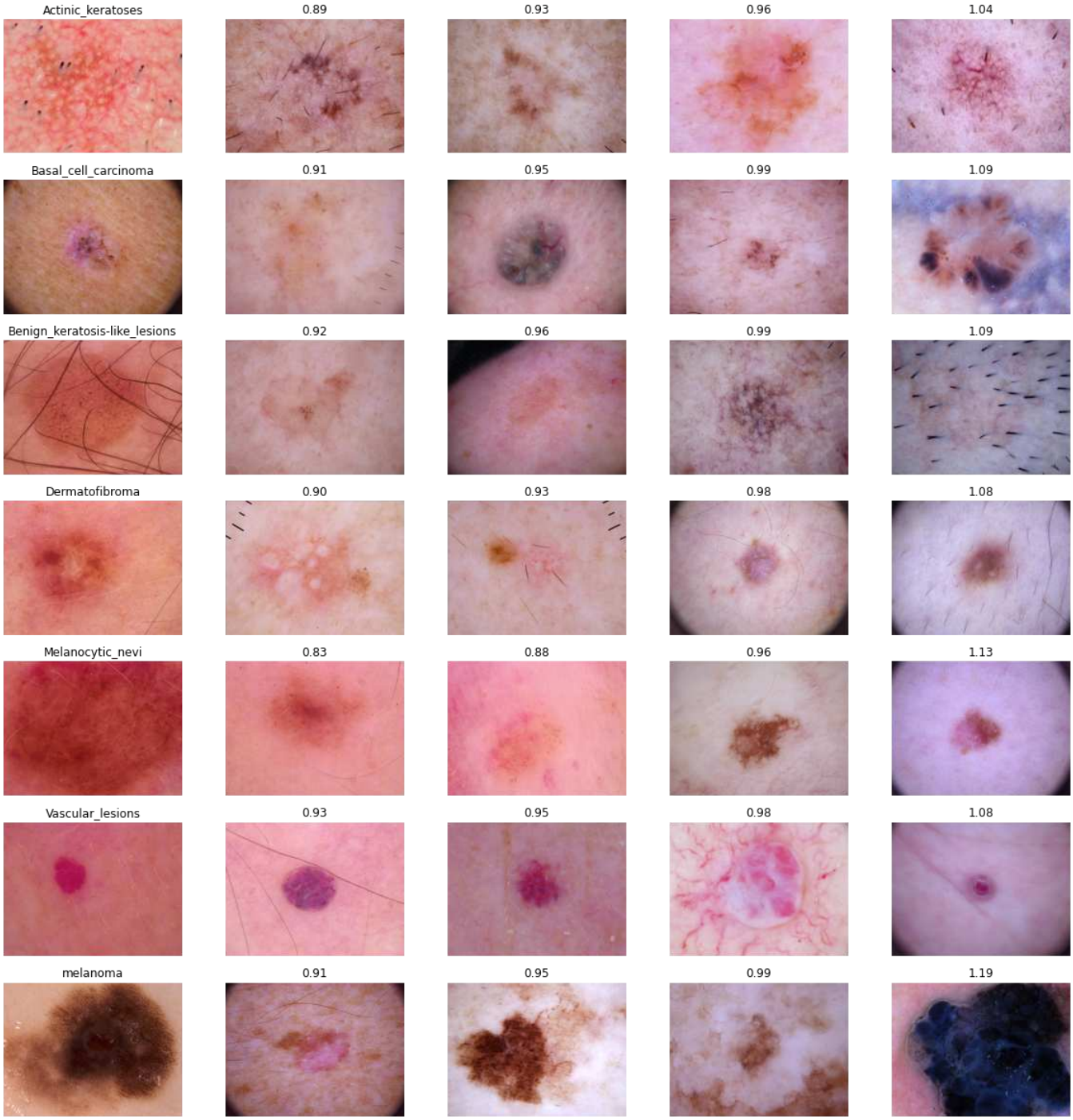
```
n_samples = 5
for sample_col in ["Red_mean", "Green_mean", "Blue_mean", "Gray_mean"]:
    fig, m_axs = plt.subplots(7, n_samples, figsize=(4 * n_samples, 3 * 7))
    fig.suptitle(f"Change in cell type appearance as {sample_col} change")
    # define a function to get back a dataframe with 5 samples sorted by color channel v
    def take_n_space(in_rows, val_col, n):
        s_rows = in_rows.sort_values([val_col])
        s_idx = np.linspace(0, s_rows.shape[0] - 1, n, dtype=int)
        return s_rows.iloc[s_idx]

    for n_axs, (type_name, type_rows) in zip(m_axs, skin_df.sort_values(["cell_type"]).g
        for c_ax, (_, c_row) in zip(n_axs, take_n_space(type_rows, sample_col, n_samples
            c_ax.imshow(c_row["image"])
            c_ax.axis("off")
            c_ax.set_title('{:2.2f}'.format(c_row[sample_col]))
        n_axs[0].set_title(type_name)
    fig.savefig("{}_samples.png".format(sample_col), dpi=300)
```

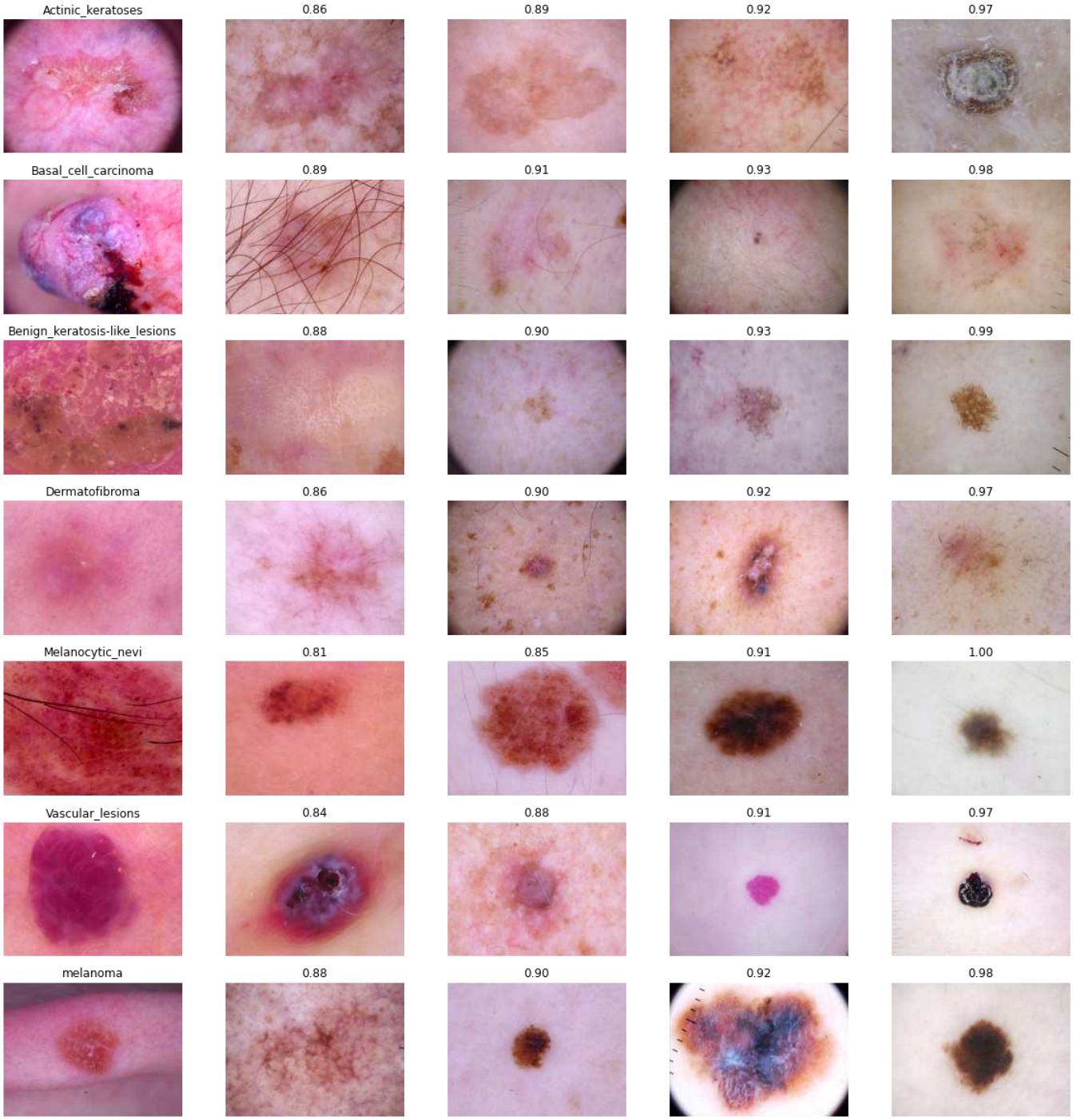
Change in cell type appearance as Red_mean change

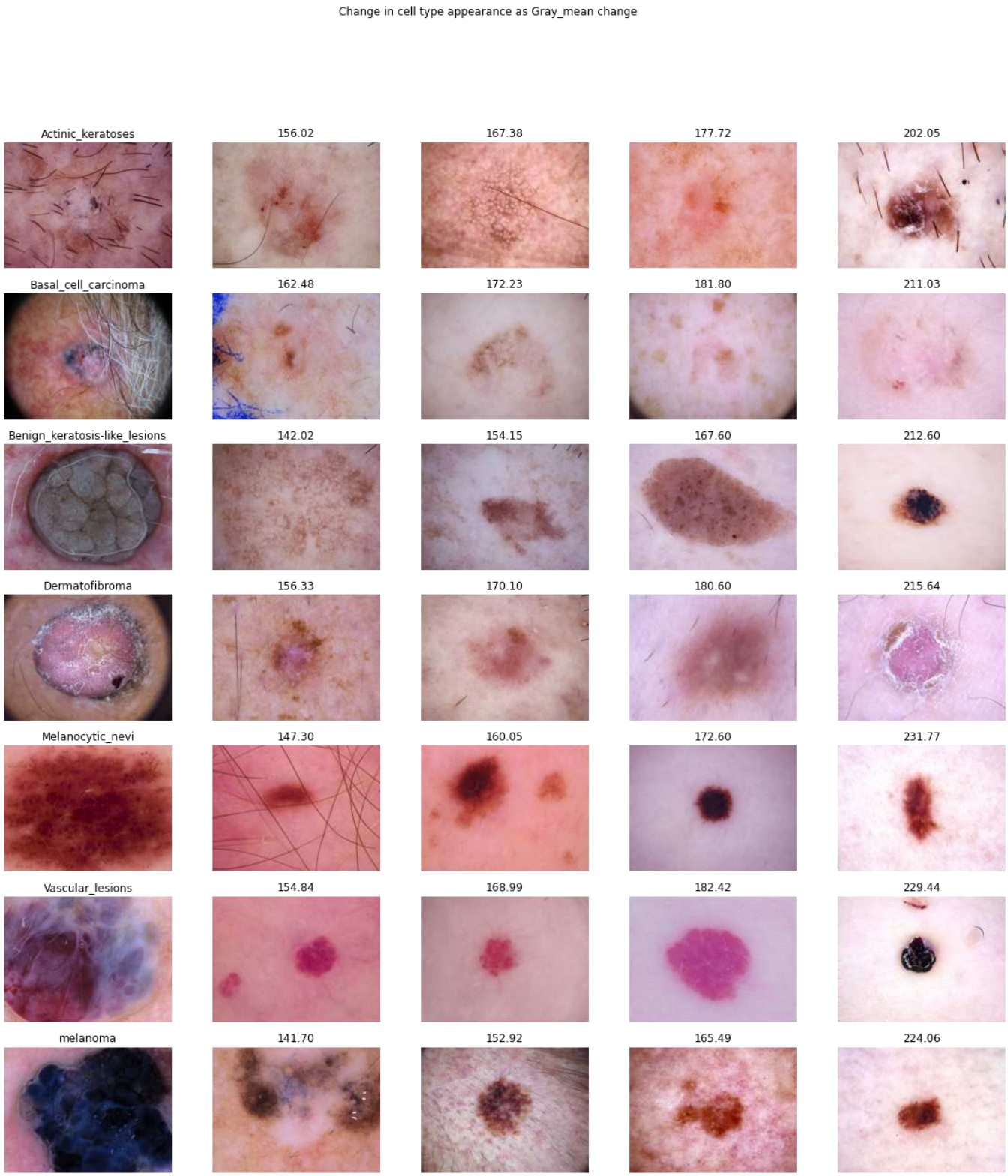


Change in cell type appearance as Green_mean change



Change in cell type appearance as Blue_mean change





Reshape image and get data for classification

In [162…

```
from PIL import Image
```

Resize image for baseline model

In [163…

```
reshaped_image = skin_df["path"].map(lambda x: np.asarray(Image.open(x).resize((64,64),
                                                                                convert("RGB")).ravel()))
```

In [164…

```
out_vec = np.stack(reshaped_image, 0)
```

In [165…
```
out_df = pd.DataFrame(out_vec)
```

In [166…
```
out_df["label"] = skin_df["cell_type_idx"]
```

In [167…
```
out_df.head()
```

Out[167]:

	0	1	2	3	4	5	6	7	8	9	…	12279	12280	12281	12282	12283	12284	12285
0	191	152	194	191	153	195	192	149	192	194	…	185	154	180	186	156	184	182
1	24	13	23	24	14	28	37	24	46	61	…	26	15	29	27	15	28	24
2	185	129	140	192	136	151	198	142	156	198	…	152	120	138	144	113	123	115
3	24	11	19	36	19	30	64	38	50	91	…	27	13	16	25	12	16	24
4	138	94	117	158	113	138	178	133	161	191	…	140	106	118	108	77	92	67

5 rows × 12289 columns

In [168…
```
out_path = "hmnist_64_64_RBG.csv"
out_df.to_csv(out_path, index=False)
```

Resize Image for Retraining Model

In [169…
```
img = Image.open(skin_df["path"][0])
img.size
```

Out[169]:
```
(600, 450)
```

In [170…
```
!mkdir skin_lesion_types
```

In [171…
```
skin_df["cell_type"].unique()
```

Out[171]:
```
array(['Benign_keratosis-like_lesions', 'Melanocytic_nevi',
       'Dermatofibroma', 'melanoma', 'Vascular_lesions',
       'Basal_cell_carcinoma', 'Actinic_keratoses'], dtype=object)
```

In [172…
```
skin_df["path"][0]
```

Out[172]:
```
'/Users/Keerthi Vardhani/Desktop/python/Skin Lesion\\skin_images\\ISIC_0027419.jpg'
```

In [ ]:
```
for index in skin_df.index.values.tolist():
    path = skin_df.iloc[index]["path"]
    cell_type_idx = skin_df.iloc[index]["cell_type"]
    img_id = skin_df.iloc[index]["image_id"]
    newpath = f"/Users/Keerthi Vardhani/Desktop/python/Skin Lesion/skin_lesion_types/{im
    img = Image.open(path)
    img = img.resize((299, 299), resample=Image.LANCZOS)
    img.save(newpath)
```

Resize Image for Keras Fine-Tuning Model

In [190…
```
reshaped_image = skin_df["path"].map(lambda x: np.asarray(Image.open(x).resize((256,192)
                                                          convert("RGB")))
```

In [193…
```
out_vec = np.stack(reshaped_image, 0)
```

In [194…
```
out_vec.shape
```

Out[194]:

```
(10015, 192, 256, 3)
```

In [195…

```
out_vec = out_vec.astype("float32")
out_vec /= 255
```

In [196…

```
labels = skin_df["cell_type_idx"].values
```

In [197…

```
from sklearn.model_selection import train_test_split
```

In [198…

```
X_train_orig, X_test, y_train_orig, y_test = train_test_split(out_vec, labels, test_size
```

In [199…

```
np.save("256_192_test.npy", X_test)
np.save("test_labels.npy", y_test)
```

In [200…

```
X_train, X_val, y_train, y_val = train_test_split(X_train_orig, y_train_orig, test_size=
```

In [201…

```
np.save("256_192_val.npy", X_val)
np.save("val_labels.npy", y_val)
```

In [202…

```
np.save("256_192_train.npy", X_train)
np.save("train_labels.npy", y_train)
```